

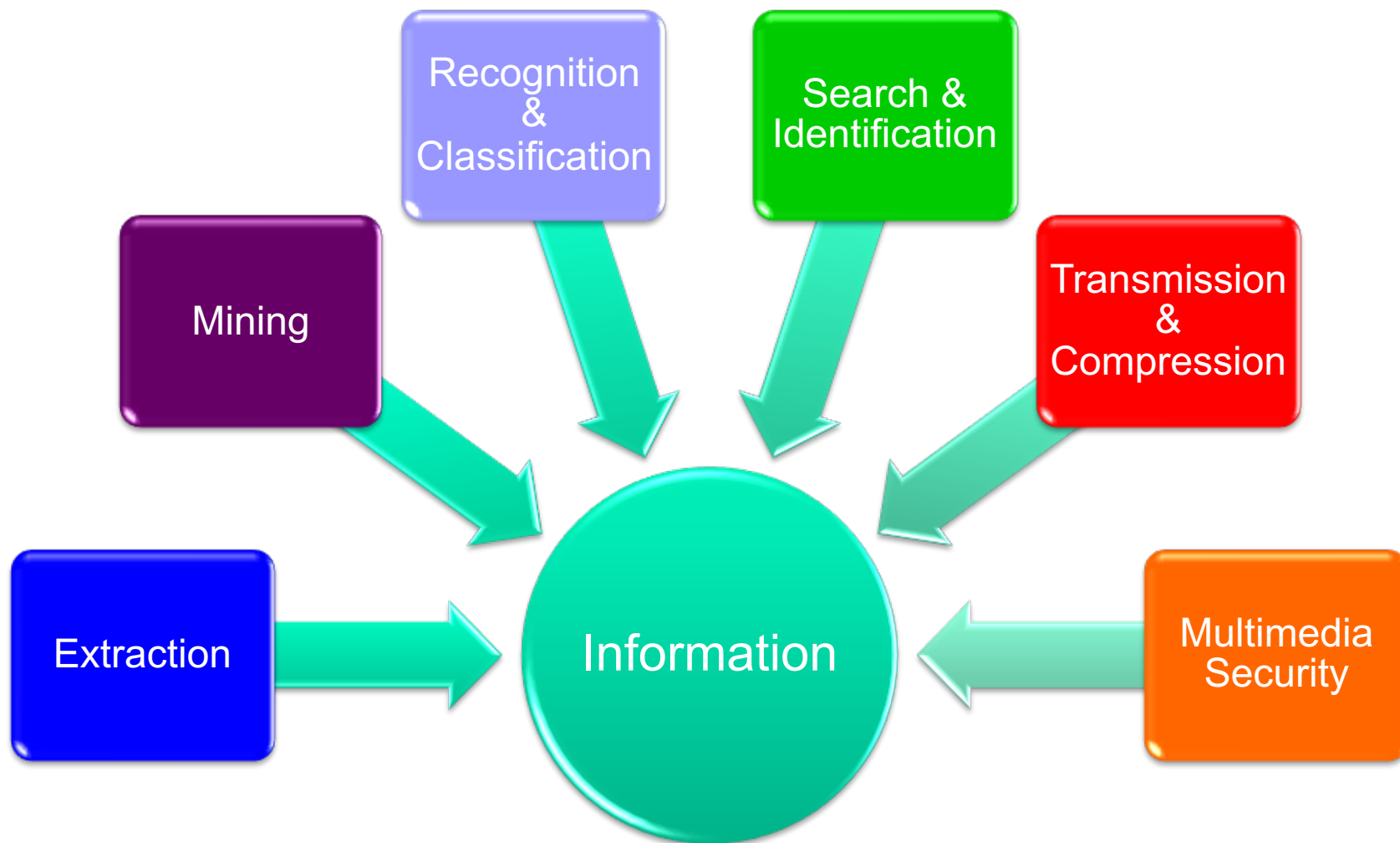
Data Science

Slava Voloshynovskiy

Outline

- Fundamental concepts and applications
- Information Theory in data science
- Detection Theory in data science

Main problems of interest



Notations

Random variables and vectors

X	Scalar random variable
$\mathbf{X}$	Random vector $\mathbf{X} = (X_1, X_2, \dots, X_N)$ or equivalent notation $\mathbf{X} = X^N$
$X_i \in \mathcal{X}$	X_i takes value in $\mathcal{X}$, where the set of possible values is: $\mathcal{X} = \{0, 1, 2, \dots\}$ for discrete variables and $\mathcal{X} = \mathbb{R}$ for continuous variables

Realization of random variables and vectors

x	Scalar variable
$\mathbf{x}$	Vector variable $\mathbf{x} = (x_1, x_2, \dots, x_N)$ or equivalent notation $\mathbf{x} = x^N$
$x_i \in \mathcal{X}$	x_i takes value in $\mathcal{X}$, where the set of possible values is: $\mathcal{X} = \{0, 1, 2, \dots\}$ for discrete variables and $\mathcal{X} = \mathbb{R}$ for continuous variables

Notations

- Basic notations of scalars and vectors

	Realization	Random
Scalar	x	X
Vector	$\mathbf{x} = x^N$	$\mathbf{X} = X^N$

$$x_i \in \mathcal{X} \quad \text{or} \quad x_i \in S_X$$

- Discrete alphabets

$$\mathcal{X} = \mathbb{Z} \quad \text{- integer}$$

$$\mathcal{X} = \{0, 1\}$$

$$\mathcal{X} = \{a, b, \dots, z\}$$

$$\mathcal{X} = \{January, February, \dots, December\}$$

- Continuous alphabets

$$\mathcal{X} = \mathbb{R}$$

- Vector $\mathbf{x} \triangleq \begin{bmatrix} | \\ \mathbf{x} \\ | \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{bmatrix} = \begin{matrix} 1 \\ 2 \\ \vdots \\ N \end{matrix} \begin{matrix} \circ \\ \circ \\ \vdots \\ \circ \end{matrix}$

$$\mathbf{X} \triangleq \begin{bmatrix} | \\ \mathbf{X} \\ | \end{bmatrix} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$$

- binary

Notations

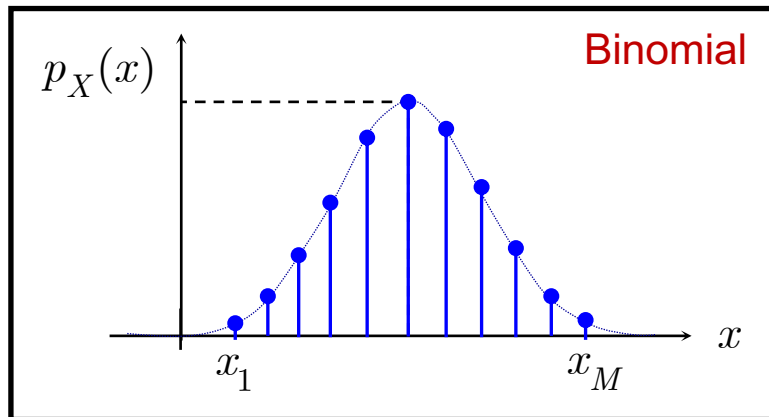
Random variables (cont)

$\Pr(\cdot)$	Probability
$p_X(x)$ (or $p(x)$)	Probability mass function (pmf) of discrete random variable X
$X \sim p_X(x)$ (or $X \sim p(x)$)	Discrete random variable X is distributed as $p_X(x)$
$p_X(x \mid Y = y)$ (or $p_X(x \mid y)$)	pmf of discrete random variable X conditioned on $Y = y$
$f_X(x)$ (or $f(x)$)	Probability density function (pdf) of continuous random variable X
$X \sim f_X(x)$ (or $X \sim f(x)$)	Continuous random variable X is distributed as $f_X(x)$
$f_X(x \mid Y = y)$ (or $f_X(x \mid y)$)	pdf of random variable X conditioned on $Y = y$

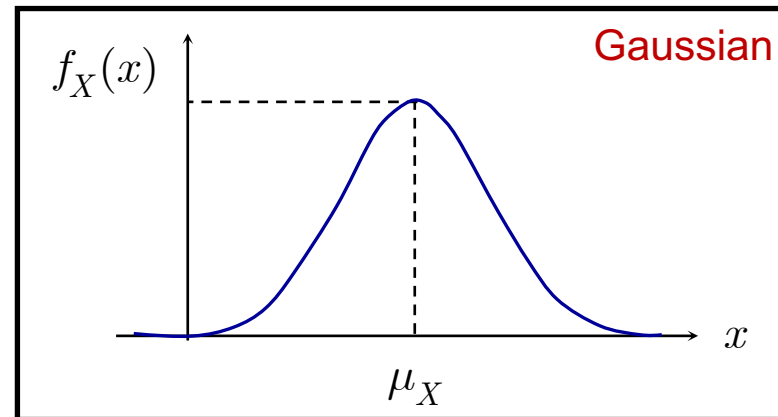
Notations

- Basic notations of probability

- Discrete random variables



- Continuous random variables



$$p_X(x) = \Pr[X = k] = \binom{M}{k} p^k (1-p)^{M-k}$$
$$k = 0, 1, \dots, M$$
$$f_X(x) = \frac{1}{\sqrt{2\pi}\sigma_X} e^{-\frac{(x-\mu_X)^2}{2\sigma_X^2}}$$

$$X \sim p_X(x) = \mathcal{B}(M, p)$$

$$X \sim f_X(x) = \mathcal{N}(\mu_X, \sigma_X^2)$$

Notations

Random variables (cont)

$\mathcal{N}(\mu, \sigma^2)$	Gaussian pdf with mean μ and variance σ^2
$X \sim \mathcal{N}(\bar{x}, \sigma_X^2)$	Scalar random variable with Gaussian pdf with mean $\bar{x}$ and variance σ_X^2
$\mathbf{X} \sim \mathcal{N}(\bar{\mathbf{x}}, \mathbf{C}_X)$	Vector random variable with Gaussian pdf with mean vector $\bar{\mathbf{x}}$ and covariance matrix $\mathbf{C}_X$
$\mathbf{X} \sim \mathcal{N}(\bar{\mathbf{x}}, \sigma_X^2 \mathbf{I}_N)$	Vector RV with Gaussian pdf with mean vector $\bar{\mathbf{x}}$ and diagonal covariance matrix of size N with σ_X^2 on the main diagonal
$\mathcal{L}(\mu, \lambda)$	Laplacian pdf with mean μ and scale λ
$\mathcal{U}(a, b)$	Uniform pdf on interval $[a, b]$
$\mathcal{GGD}(\mu, \gamma, \lambda)$	Generalized Gaussian pdf with mean μ , shape parameter γ and scale λ
$\eta(\gamma, \lambda)$	Parameter of $\mathcal{GGD}$ pdf

Notations

Random variables (cont)

$E_X[\cdot]$ Expected value taken over RV X . When there is no possible confusion, $E[\cdot]$ can be written.

$E[\cdot \mid \cdot]$ Conditional expected value

$\text{Var}[\cdot]$ Variance

$\text{Std}[\cdot]$ Standard deviation

σ_X^2 Variance of random variable X

$*$ Convolution operator

$\langle \cdot, \cdot \rangle$ Scalar product operator

Notations

▪ Moments

▪ Discrete random variables

1st moment
(mean)

$$\mu_X = E_{p_X(x)}[X] = \sum_{i=1}^M xp(x_i)$$

2nd moment
(variance)

$$\begin{aligned}\sigma_X^2 &= Var[X] = E_{p_X(x)}[|X - \mu_X|^2] \\ &= \sum_{i=1}^M (x_i - \mu_X)^2 p(x_i)\end{aligned}$$

3rd moment
(skewness)

$$E_{p_X(x)}[|X - \mu_X|^3] / \sigma_X^3$$

4th moment
(kurtosis)

$$E_{p_X(x)}[|X - \mu_X|^4] / \sigma_X^4$$

▪ Continuous random variables

$$\mu_X = E_{f_X(x)}[X] = \int_{x \in \mathcal{X}} xf_X(x) dx$$

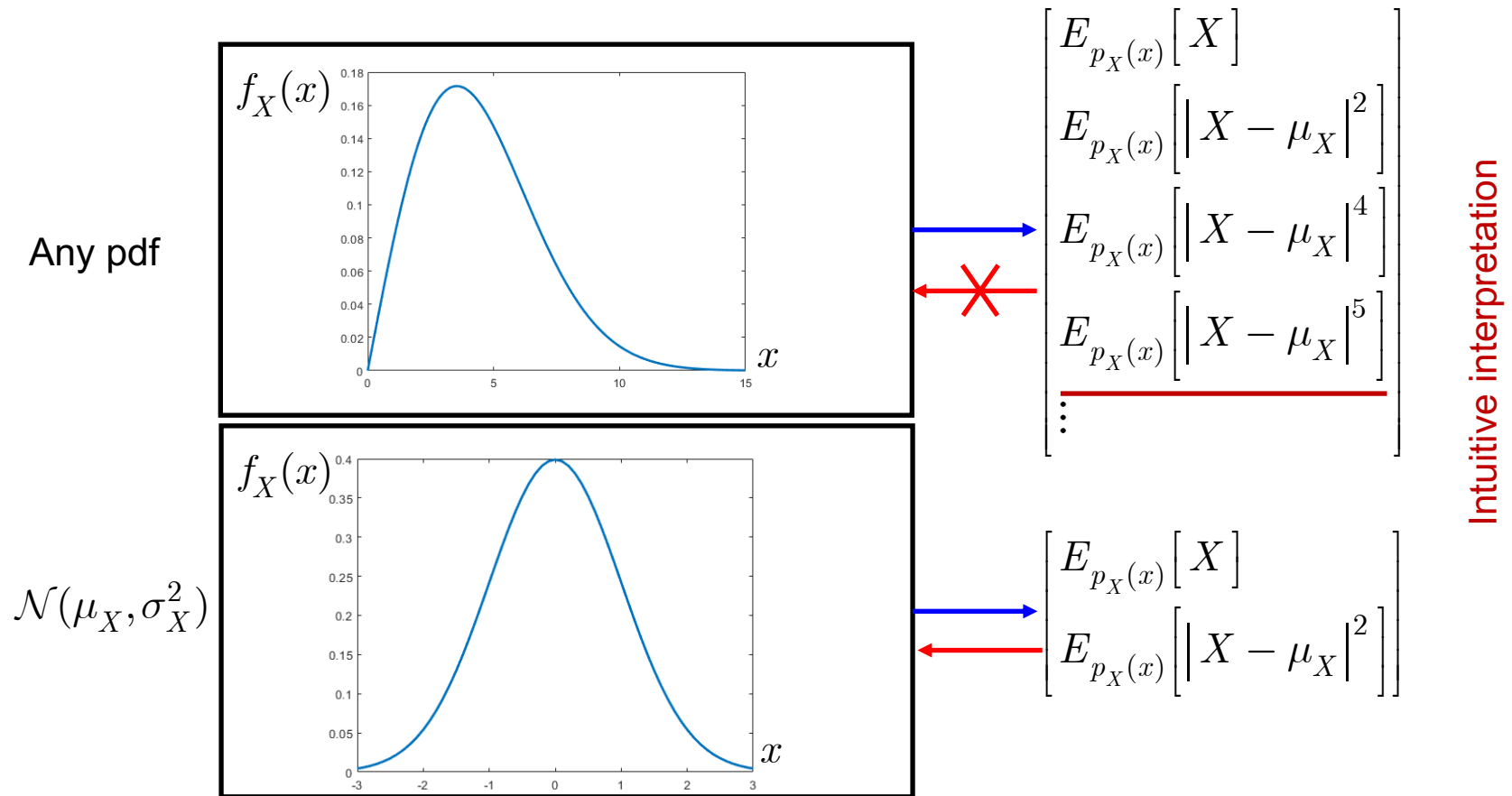
$$\begin{aligned}\sigma_X^2 &= Var[X] = E_{f_X(x)}[|X - \mu_X|^2] \\ &= \int_{x \in \mathcal{X}} (x - \mu_X)^2 f_X(x) dx\end{aligned}$$

$$E_{f_X(x)}[|X - \mu_X|^3] / \sigma_X^3$$

$$E_{f_X(x)}[|X - \mu_X|^4] / \sigma_X^4$$

Notations

- General case: given a pmf/pdf, we can compute moments (not for all)



Notations

- **Characteristic function** of pdf is defined as:

$$\Phi(t) = \int_{-\infty}^{\infty} f(x)e^{jtx} dx = \mathbb{E}_{f(x)} [e^{jtX}]$$

where $j = \sqrt{-1}$

- The pdf can be recovered as:

$$f(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \Phi(t)e^{-jtx} dt$$

Moment-generating function of pdf:

$$M(t) = \mathbb{E}_{f(x)} [e^{tX}]$$

Notations

- Taylor's series expansion:

$$\exp(tX) = \sum_0^{\infty} \frac{(t)^n \cdot X^n}{n!}$$

$$\Phi(t) = \mathbb{E}_{f(x)}[\exp(jtX)] = 1 + \frac{jt\mathbb{E}_{f(x)}[X]}{1} - \frac{t^2\mathbb{E}_{f(x)}[X^2]}{2!} + \dots + \frac{(jt)^n\mathbb{E}_{f(x)}[X^n]}{n!}$$
$$M(t) = \mathbb{E}_{f(x)}[\exp(tX)] = 1 + \frac{t\mathbb{E}_{f(x)}[X]}{1} + \frac{t^2\mathbb{E}_{f(x)}[X^2]}{2!} + \dots + \frac{t^n\mathbb{E}_{f(x)}[X^n]}{n!}$$

The first four moments define the mean, variance, skewness, and kurtosis of the PDF $f(x)$, respectively.

Notations

- Practical computation of moments from i.i.d. sequence $\mathbf{x} = (x_1, \dots, x_N)$ drawn from $f(x)$

- The nth **empirical pdf moment estimation**

$$\hat{m}_n = \frac{1}{N} \sum_{i=1}^N x_i^n, \quad n \geq 1$$

- Which is an unbiased estimation of **true pdf moments**

$$m_n = \mathbb{E}[X^n] = \int_{-\infty}^{\infty} f(x) x^n dx$$

- **Moments of CDF:**

$$\mathbf{M}_n = \int_{-\infty}^{\infty} \Phi(t) t^n dt$$

$$\text{Link to pdf: } \mathbf{M}_n = j^n 2\pi \frac{d^n}{dx^n} f(x) \Big|_{x=0}$$

Notations

- Practical computation of moments from i.i.d. sequence $\mathbf{x} = (x_1, \dots, x_N)$ drawn from $f(x)$

- **Empirical moments of CDF:**

- Estimate empirical M-bin histogram $\{h(m)\}_{m=0}^{M-1}$ of pdf $f(x)$
- Estimate K-point CDF with $K = 2^{\lceil \log_2 M \rceil}$ as

$$\Phi(k) = \sum_{m=0}^{M-1} h(m) \exp \left\{ \frac{j2\pi mk}{K} \right\}$$

Note that Fourier works in two ways:

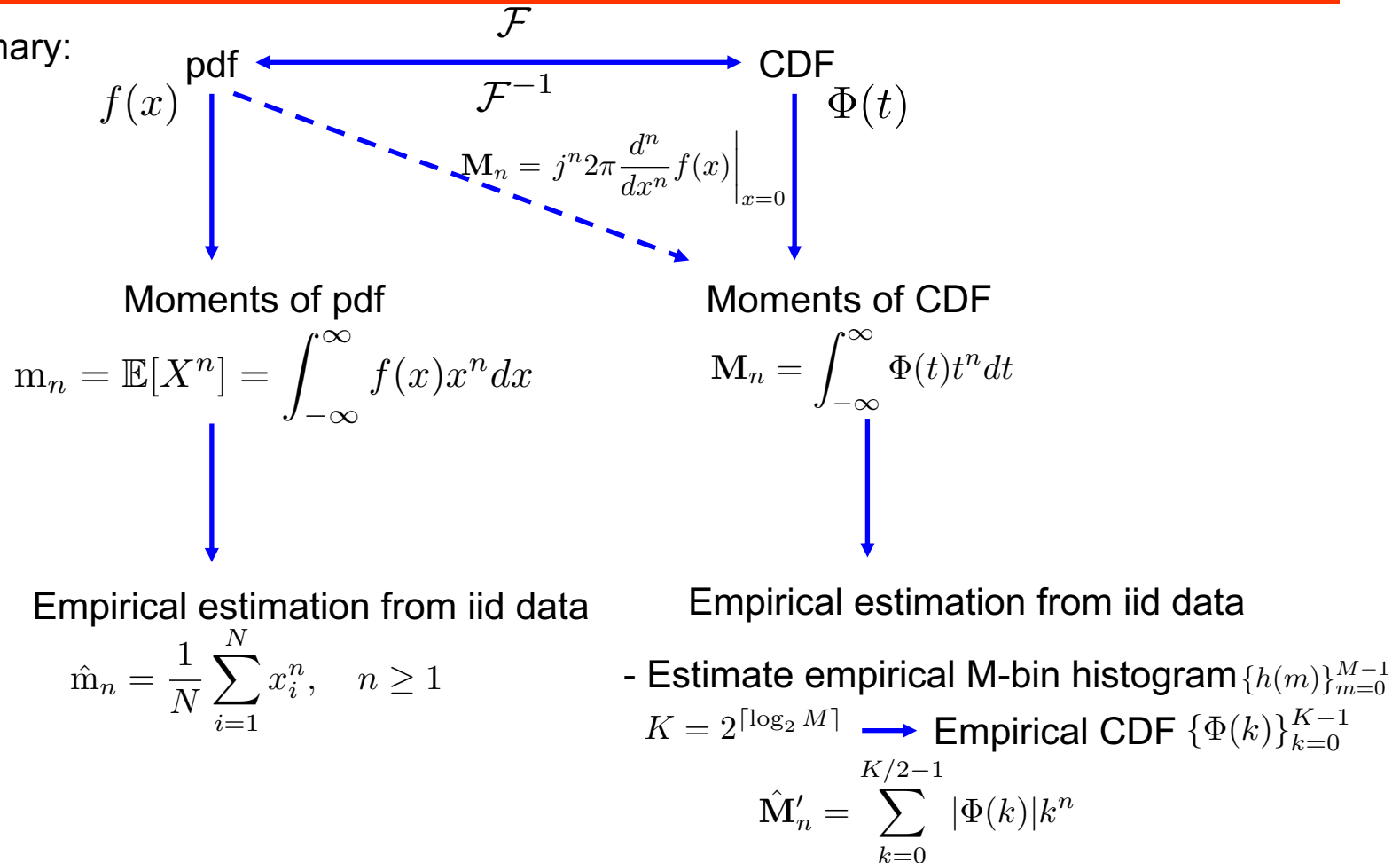
$$h(m) = \frac{1}{K} \sum_{k=0}^{K-1} \Phi(k) \exp \left\{ -\frac{j2\pi mk}{K} \right\}$$
$$0 \leq m \leq M - 1$$

- Compute empirical CDF moments as:

$$\hat{\mathbf{M}}'_n = \sum_{k=0}^{K/2-1} |\Phi(k)| k^n \qquad \hat{\mathbf{M}}_n = \sum_{k=0}^{K-1} \Phi(k) \sin^n \left(\frac{\pi k}{K} \right)$$

Notations

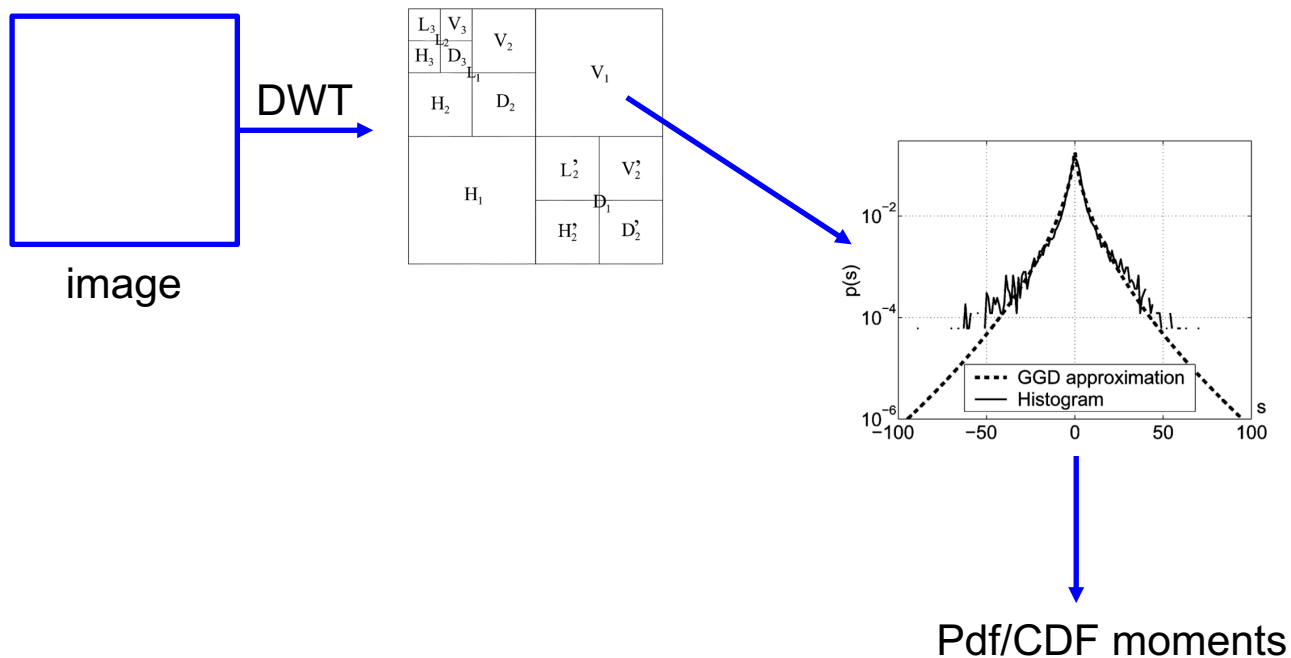
■ Summary:



Notations

Applications:

- Outlier detection
- Fake detection
- Secret message detection in images (steganalysis)



Notations

- Properties of variance and covariance
 - Let X and Y be random variables and a, b, c be constants.

$$\text{Var}[X] = E\left[\|X - E[X]\|^2\right] = E[X^2] - E[X]^2$$

$$0 \leq \text{Var}[X] \leq E[X^2]$$

$$\text{Var}[cX] = c^2 \text{Var}[X]$$

$$\text{Var}[aX + bY] = a^2 \text{Var}[X] + 2ab \text{Cov}[X, Y] + b^2 \text{Var}[Y]$$

$$\text{Cov}[X, Y] = E_{p(x,y)}[(X - \mu_X)(Y - \mu_Y)] = \sum_{(x,y) \in S_{X,Y}} p(x,y)(x - \mu_X)(y - \mu_Y)$$

$$= E_{p(x,y)}[XY] - \mu_X \mu_Y = \sigma_{XY}$$

Notations

■ Vector $\mathbf{x} \triangleq \begin{bmatrix} | \\ \mathbf{x} \\ | \end{bmatrix} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_N \end{bmatrix} \xrightarrow{\text{random}} \mathbf{X} \triangleq \begin{bmatrix} | \\ \mathbf{X} \\ | \end{bmatrix} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$

$$\mathbf{X} \triangleq \begin{bmatrix} | \\ \mathbf{X} \\ | \end{bmatrix} = \begin{bmatrix} X_1 \\ \color{red}{X_i} \\ \vdots \\ X_N \end{bmatrix}$$

$\color{red}{p_X(x_i)}$
 $\color{red}{f_X(x_i)}$

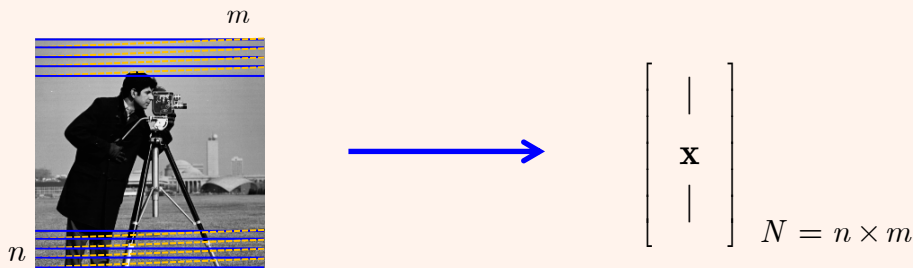
- Joint pmf or pdf

$$p_{\mathbf{x}}(\mathbf{x}) = p_{\mathbf{x}}(x_1, x_2, \dots, x_N)$$

$$f_{\mathbf{x}}(\mathbf{x}) = f_{\mathbf{x}}(x_1, x_2, \dots, x_N)$$

Notations

- **Example 1:**
 - Consider an image as a matrix
 - Read in a zig-zag order into one vector

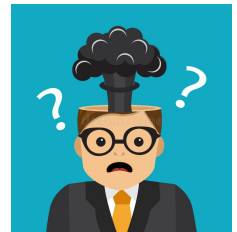


- Joint pmf defines **all statistical relations** between all elements of image/vector

$$\mathbf{X} = \begin{bmatrix} | \\ \mathbf{X} \\ | \end{bmatrix} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$$

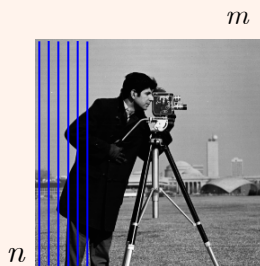
- Complexity

- Modern image of size 1000x1000
 $N = n \times m = 1000 \times 1000 = 1000000$
All pair-wise, triple-wise, etc.



Notations

- **Example 2:**
 - Consider an image as a matrix
 - Read by rows as a matrix

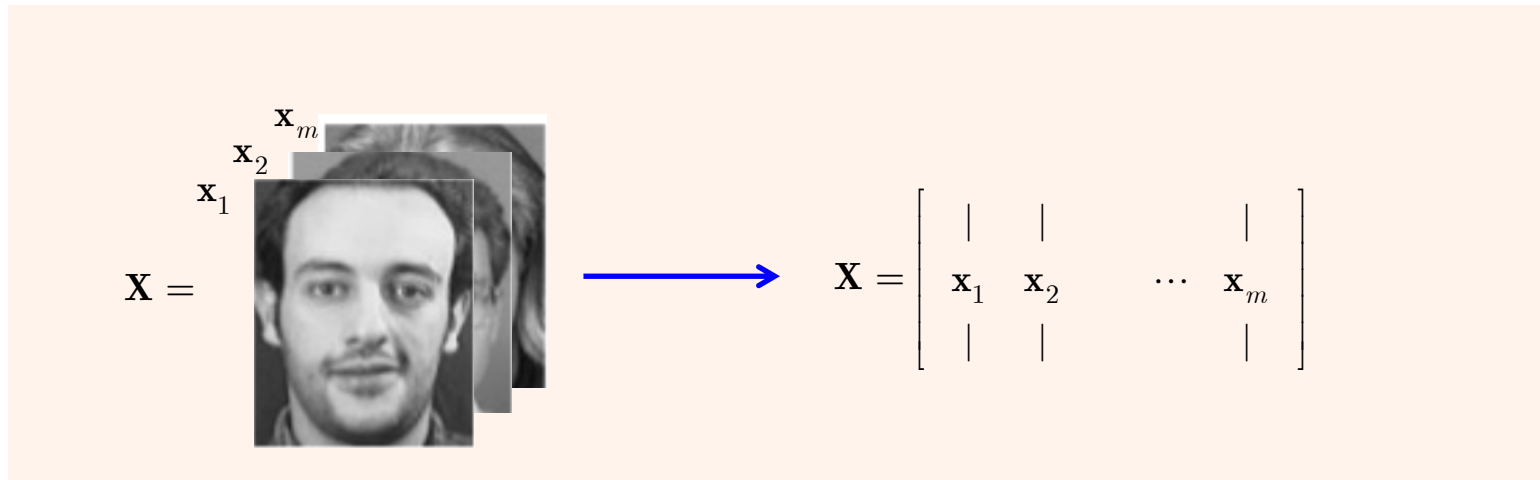


$$\longrightarrow \mathbf{X} = \begin{bmatrix} | & | & & | \\ \mathbf{x}_1 & \mathbf{x}_2 & \cdots & \mathbf{x}_m \\ | & | & & | \end{bmatrix}$$

n

Notations

- **Example 3:**
 - image represents a column in a data matrix



Random vectors: chain rule decomposition

- Chain rule for probability: joint probability

$$p_{\mathbf{x}}(\mathbf{x}) = p_{\mathbf{x}}(x_1, x_2, \dots, x_N) = p(x_1, x_2, \dots, x_N) \quad (\text{Notations})$$

$$\begin{aligned} p(x_1, x_2, \dots, x_N) &= p(x_1) p(x_2 | x_1) p(x_3 | x_2, x_1) \dots p(x_N | x_{N-1}, \dots, x_2, x_1) = \\ &= \prod_{i=1}^N p(x_i | x_{i-1}, \dots, x_2, x_1) \end{aligned}$$

$$p(x_1, x_2) = p(x_1) p(x_2 | x_1) = p(x_2) p(x_1 | x_2)$$

$$\Rightarrow p(x_2 | x_1) = \frac{p(x_2) p(x_1 | x_2)}{p(x_1)}$$

Random vectors: independence

- Using the chain rule

$$\begin{aligned} p(x_1, x_2, \dots, x_N) &= p(x_1) p(x_2 | x_1) p(x_3 | x_2, x_1) \dots p(x_N | x_{N-1}, \dots, x_2, x_1) = \\ &= \prod_{i=1}^N p(x_i | x_{i-1}, \dots, x_2, x_1) \end{aligned}$$

$$p(x_1, x_2) = p(x_1) p(x_2 | x_1) = p(x_2) p(x_1 | x_2) \Rightarrow p(x_1) p(x_2)$$

independent
 $X_1 \perp X_2$

$$p_{\mathbf{x}}(x_1, x_2, \dots, x_N) = p_{x_1}(x_1) p_{x_2}(x_2) p_{x_3}(x_3) \dots p_{x_N}(x_N) = \prod_{i=1}^N p_{x_i}(x_i)$$

For the **independent and identically distributed (i.i.d.)** random variables

$$p_{\mathbf{x}}(x_1, x_2, \dots, x_N) = p_x(x_1) p_x(x_2) p_x(x_3) \dots p_x(x_N) = \prod_{i=1}^N p_x(x_i)$$

Random vectors: independence

- Practical application of independence

$$p_{\mathbf{x}}(\mathbf{x}) = p_{\mathbf{x}}(x_1, x_2, \dots, x_N)$$

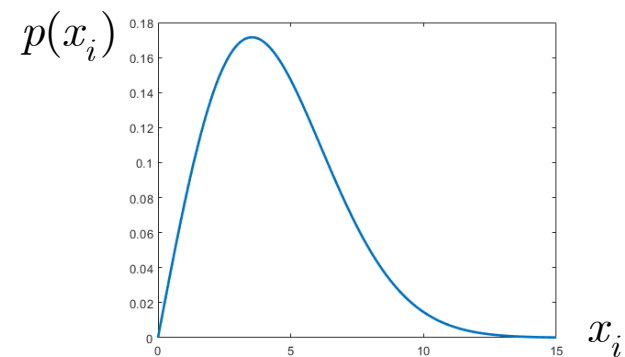


$$\mathbf{X} \triangleq \begin{bmatrix} | \\ \mathbf{X} \\ | \end{bmatrix} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$$

Simplification

$$p(x_1, x_2, \dots, x_N) = p(x_1)p(x_2)p(x_3)\dots p(x_N)$$

$$\mathbf{X} \triangleq \begin{bmatrix} | \\ \mathbf{X} \\ | \end{bmatrix} = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix} \sim \begin{matrix} p(x_1) \\ p(x_2) \\ p(x_N) \end{matrix}$$



Random vectors: Markovianity

- The **first order** Markov model
 - assumes the dependence on **one element** from past

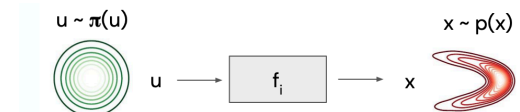
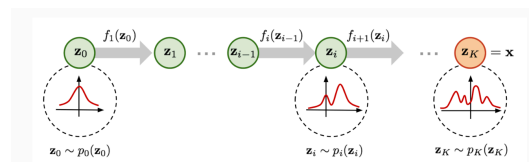
$$p(x_i | x_{i-1}, x_{i-2}, \dots, x_1) \xRightarrow{\text{1st order Markov model}} p(x_i | x_{i-1})$$

$$p(x_1, x_2, \dots, x_N) = p(x_1) p(x_2 | x_1) p(x_3 | x_2, x_1) \dots p(x_N | x_{N-1}, \dots, x_2, x_1) =$$

$$\xRightarrow{\text{1st order Markov model}} p(x_1) p(x_2 | x_1) p(x_3 | x_2) \dots p(x_N | x_{N-1})$$

$$p(x_1, x_2, \dots, x_N) \xRightarrow{\text{1st order Markov model}} p(x_1) \prod_{i=2}^N p(x_i | x_{i-1})$$

The modeling of a complex pmf/pdf is reduced to the definition of pair-wise conditional distributions



HAMILTONIAN GENERATIVE NETWORKS

Random vectors: Markovianity

- The **second order** Markov model
 - assumes the dependence on **two elements** from past

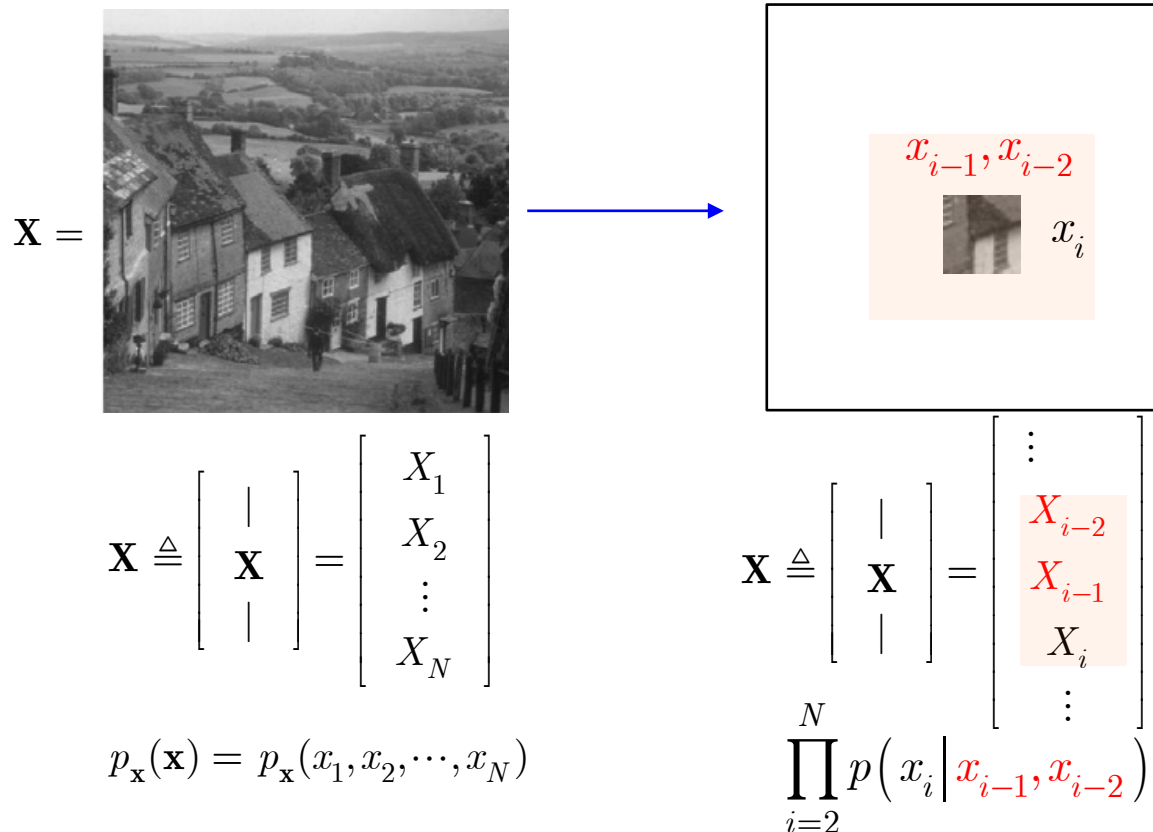
$$p(x_i | x_{i-1}, x_{i-2}, \dots, x_1) \xRightarrow{\text{2nd order Markov model}} p(x_i | x_{i-1}, x_{i-2})$$

$$p(x_1, x_2, \dots, x_N) = p(x_1) p(x_2 | x_1) p(x_3 | x_2, x_1) \dots p(x_N | x_{N-1}, \dots, x_2, x_1) =$$
$$\xRightarrow{\text{2nd order Markov model}} p(x_1) p(x_2 | x_1) p(x_3 | x_2, x_1) \dots p(x_N | x_{N-1}, x_{N-2})$$

$$p(x_1, x_2, \dots, x_N) \xRightarrow{\text{2nd order Markov model}} p(x_1) \prod_{i=2}^N p(x_i | x_{i-1}, x_{i-2})$$

Random vectors: Markovianity

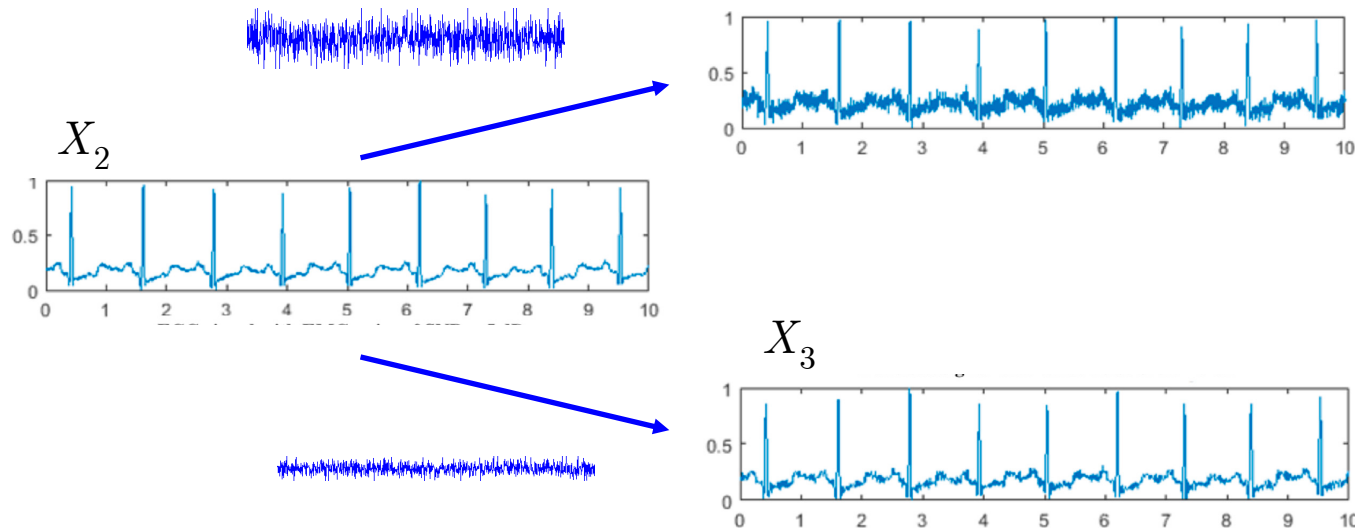
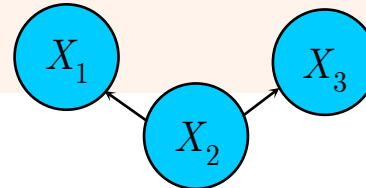
- Practical application application of Markovianity for image processing



Random vectors: conditional independence

- Random variables X_1 and X_3 are independent given X_2

$$p(x_1, x_2, x_3) = p(x_2) \underbrace{p(x_1, x_3 | x_2)}_{=p(x_1|x_2)p(x_3|x_2)} = p(x_2)p(x_1|x_2)p(x_3|x_2)$$



Notations

Vector norms

$$\mathbf{x} = (x_1, x_2, \dots, x_N)$$

$$\mathbf{x} \in \mathbb{R}^N$$

$$\|\mathbf{x}\|_p$$

L_p - norm or p - norm

$$\|\mathbf{x}\|_p = \left(|x_1|^p + |x_2|^p + \dots + |x_N|^p \right)^{\frac{1}{p}} = \left(\sum_{i=1}^N |x_i|^p \right)^{\frac{1}{p}}$$

$$\|\mathbf{x}\|_2$$

Euclidian norm

$$\|\mathbf{x}\|_2 = \left(|x_1|^2 + |x_2|^2 + \dots + |x_N|^2 \right)^{\frac{1}{2}} = \left(\sum_{i=1}^N |x_i|^2 \right)^{\frac{1}{2}}$$

$$\|\mathbf{x}\|_1$$

L_1 - norm

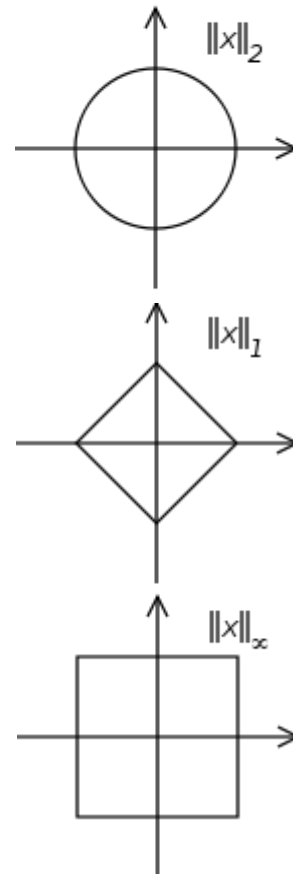
$$\|\mathbf{x}\|_1 = (|x_1| + |x_2| + \dots + |x_N|) = \left(\sum_{i=1}^N |x_i| \right)$$

$$\|\mathbf{x}\|_\infty$$

L_∞ - norm or maximum norm

$$\|\mathbf{x}\|_\infty = \max \{ |x_1|, |x_2|, \dots, |x_N| \}$$

Unit-norm



Notations

Distances and similarity measures

$$\|\mathbf{x} - \mathbf{y}\|_2$$

Euclidian distance between two vectors

$$d_2(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_2 = \left(\sum_{i=1}^N (x_i - y_i)^2 \right)^{\frac{1}{2}}$$

$$\|\mathbf{x} - \mathbf{y}\|_1$$

L_1 - distance between two vectors

$$d_1(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\|_1 = \sum_{i=1}^N |x_i - y_i|$$

$$\mathbf{x} \oplus \mathbf{y}$$

Hamming distance between two vectors

$$d^H(\mathbf{x}, \mathbf{y}) = \mathbf{x} \oplus \mathbf{y} = \sum_{i=1}^N 1\{x_i, y_i\}$$

$$\text{with indicator function } 1\{x_i, y_i\} = \begin{cases} 0, & \text{if } x_i = y_i \\ 1, & \text{if } x_i \neq y_i \end{cases}$$

Notations

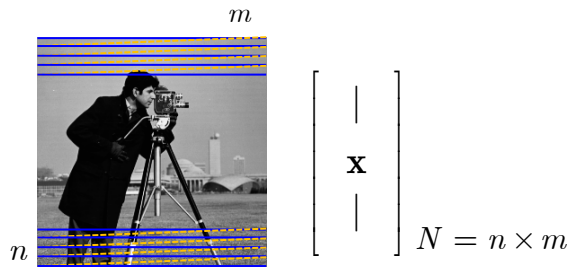
Distances and similarity measures for random vectors

$$D = E_{f(\mathbf{x})}[d(\mathbf{X}, \mathbf{Y})]$$

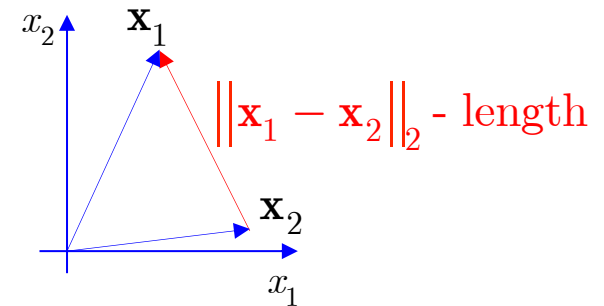
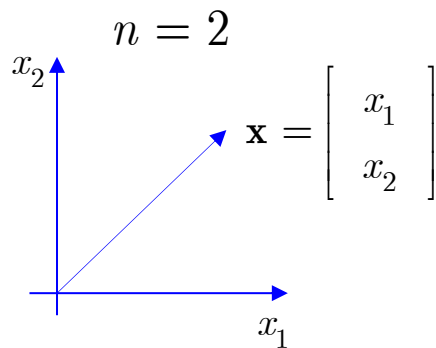
Expected distance between two random vectors with pdf $f(\mathbf{x})$

$$d(\mathbf{X}, \mathbf{Y}) = \frac{1}{N} \sum_{i=1}^N d(X_i, Y_i)$$

Distances

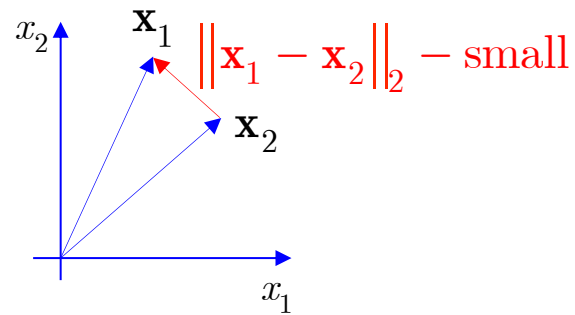
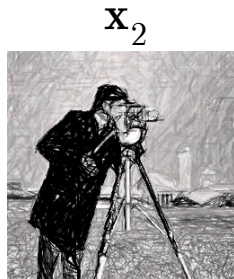
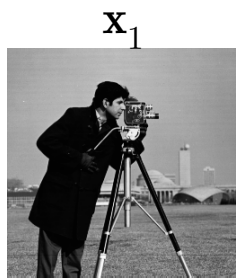


$$d_2(\mathbf{x}_1, \mathbf{x}_2) = \|\mathbf{x}_1 - \mathbf{x}_2\|_2 = \left(\sum_{i=1}^N (x_{1i} - x_{2i})^2 \right)^{\frac{1}{2}}$$

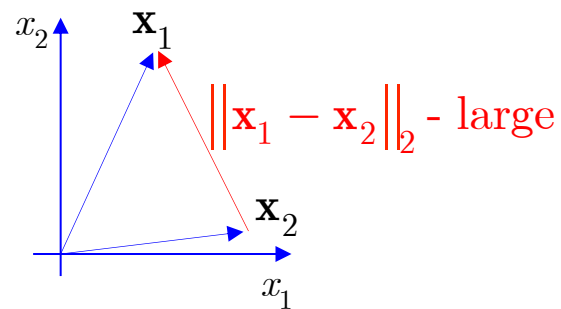
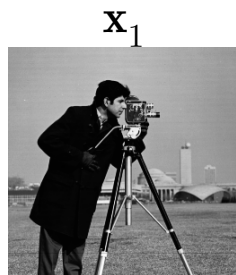


Distances

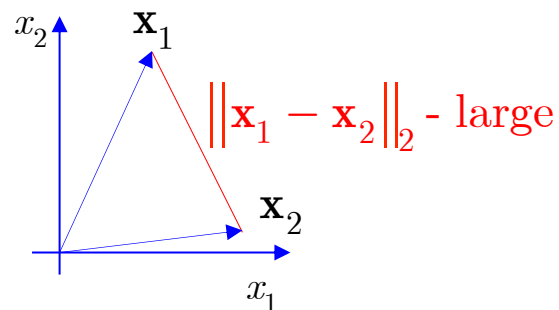
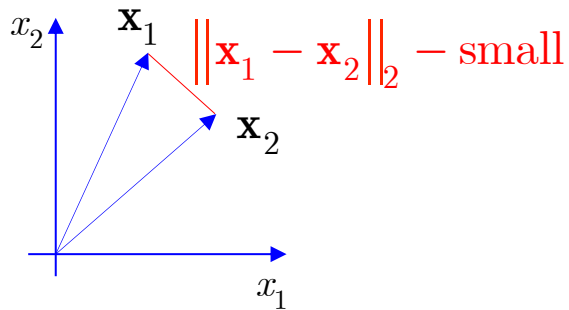
- Perceptually close images



- Perceptually different images



Distances: link to inner product



$$\|\mathbf{x}_1 - \mathbf{x}_2\|_2^2 = \|\mathbf{x}_1\|_2^2 - 2 \underbrace{\mathbf{x}_1^T \mathbf{x}_2}_{\text{inner product}} + \|\mathbf{x}_2\|_2^2$$

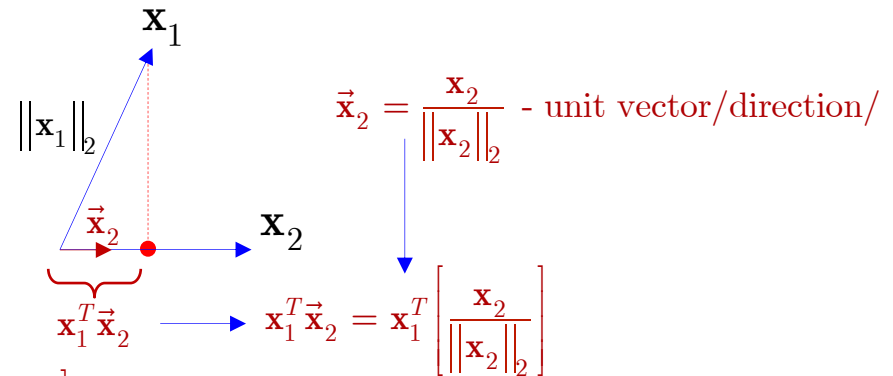
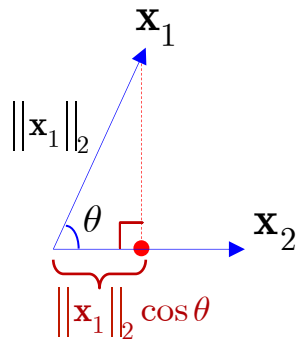
Recall B: matrix operations

$$\mathbf{x}^T \mathbf{y} = \mathbf{y}^T \mathbf{x} \in \mathbb{R} \triangleq \begin{bmatrix} x_1 & x_2 & \cdots & x_n \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} - & \mathbf{x} & - \end{bmatrix} \begin{bmatrix} | \\ \mathbf{y} \\ | \end{bmatrix} = \sum_{i=1}^n x_i y_i$$

$$\begin{aligned} \|\mathbf{x}_1 - \mathbf{x}_2\|_2 - \text{small} &\Rightarrow \mathbf{x}_1^T \mathbf{x}_2 - \text{large} \\ \|\mathbf{x}_1 - \mathbf{x}_2\|_2 - \text{large} &\Rightarrow \mathbf{x}_1^T \mathbf{x}_2 - \text{small} \end{aligned}$$

Distances: link to cosine similarity

- Cosine similarity



$$\|\mathbf{x}_1\|_2 \cos \theta = \mathbf{x}_1^T \vec{\mathbf{x}}_2 = \mathbf{x}_1^T \left[\frac{\mathbf{x}_2}{\|\mathbf{x}_2\|_2} \right]$$

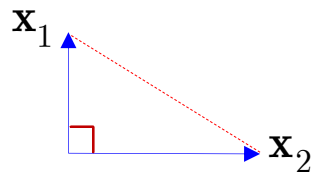
$$\mathbf{x}_1^T \mathbf{x}_2 = \|\mathbf{x}_1\|_2 \|\mathbf{x}_2\|_2 \cos \theta$$

$$\Rightarrow \cos \theta = \frac{\mathbf{x}_1^T \mathbf{x}_2}{\|\mathbf{x}_1\|_2 \|\mathbf{x}_2\|_2}$$

Distances: orthogonal and collinear vectors

$$\|\mathbf{x}_1 - \mathbf{x}_2\|_2^2 = \|\mathbf{x}_1\|_2^2 + \|\mathbf{x}_2\|_2^2 - 2\mathbf{x}_1^T \mathbf{x}_2 = \|\mathbf{x}_1\|_2^2 + \|\mathbf{x}_2\|_2^2 - 2\cos\theta \|\mathbf{x}_1\|_2 \|\mathbf{x}_2\|_2$$

Orthogonal vectors

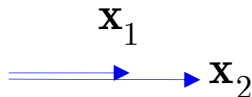


$$\mathbf{x}_1^T \mathbf{x}_2 = 0$$

$$\|\mathbf{x}_1\|_2 \|\mathbf{x}_2\|_2 \underbrace{\cos 90^\circ}_{=0} = 0$$

$$\|\mathbf{x}_1 - \mathbf{x}_2\|_2^2 = \|\mathbf{x}_1\|_2^2 + \|\mathbf{x}_2\|_2^2 - 2\underbrace{\mathbf{x}_1^T \mathbf{x}_2}_{=0} = \|\mathbf{x}_1\|_2^2 + \|\mathbf{x}_2\|_2^2$$

Collinear vectors



$$\mathbf{x}_1^T \mathbf{x}_2 = \max$$

$$\|\mathbf{x}_1\|_2 \|\mathbf{x}_2\|_2 \underbrace{\cos 0^\circ}_{=1} = \|\mathbf{x}_1\|_2 \|\mathbf{x}_2\|_2$$

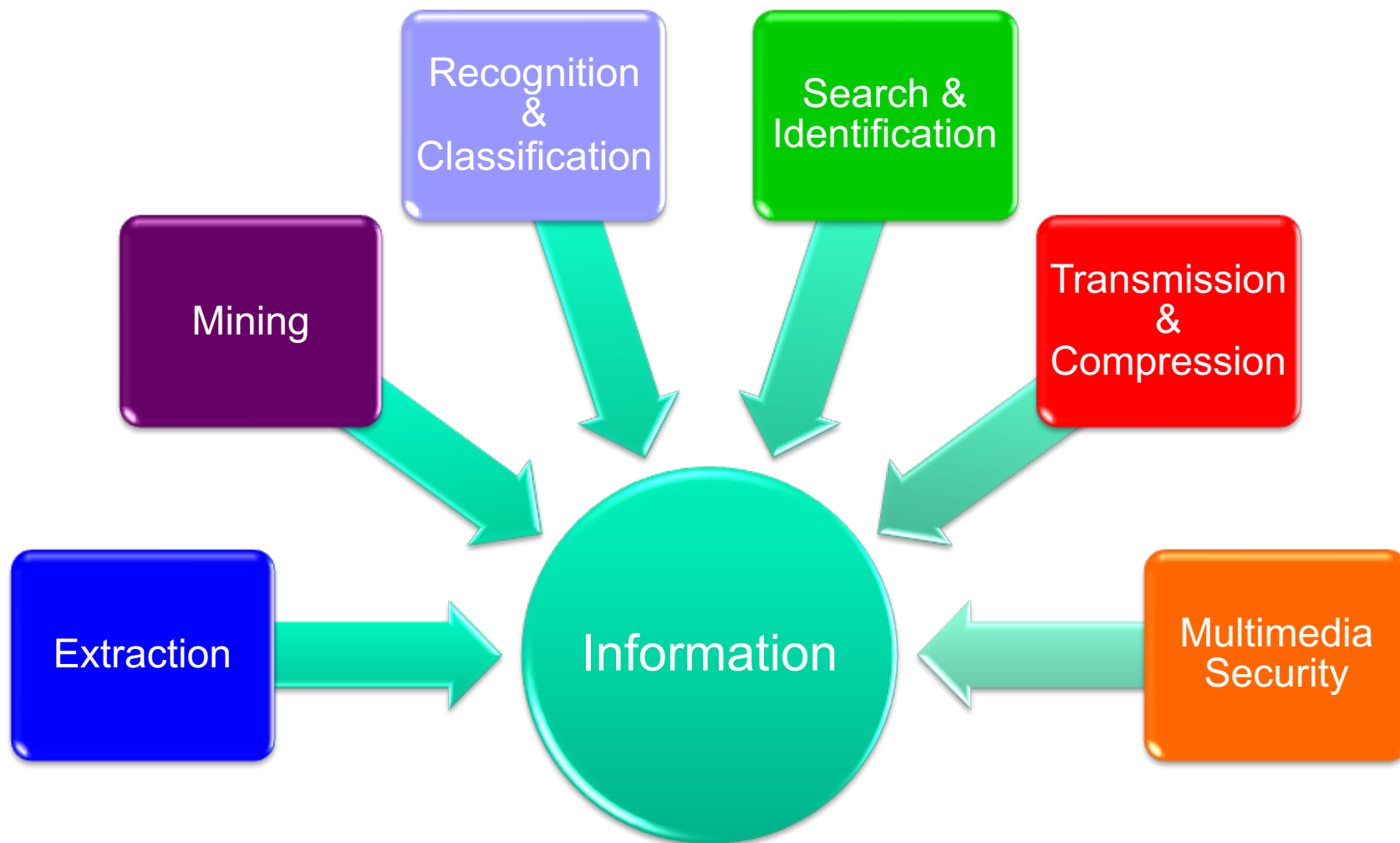
$$\|\mathbf{x}_1 - \mathbf{x}_2\|_2^2 = \|\mathbf{x}_1\|_2^2 + \|\mathbf{x}_2\|_2^2 - 2\|\mathbf{x}_1\|_2 \|\mathbf{x}_2\|_2$$

$$\mathbf{x}_1 = \mathbf{x}_2$$

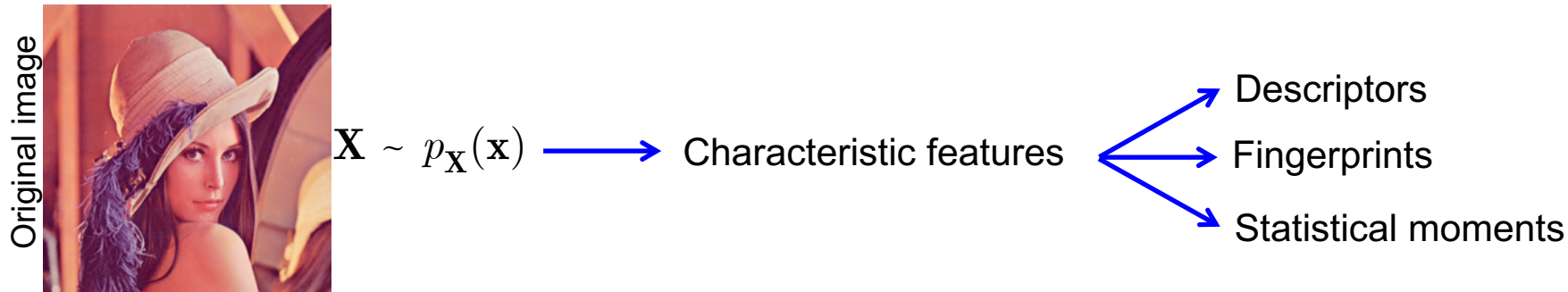
$$\|\mathbf{x}_1\|_2 = \|\mathbf{x}_2\|_2$$

$$\|\mathbf{x}_1 - \mathbf{x}_2\|_2^2 = 0$$

Main problems of interest



Extraction

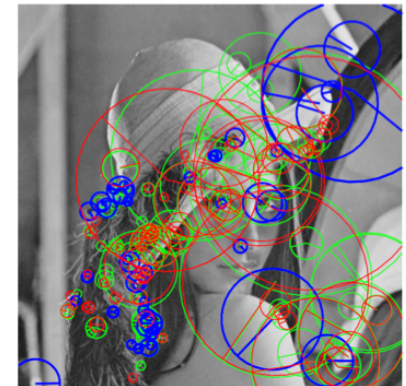


Extraction of features

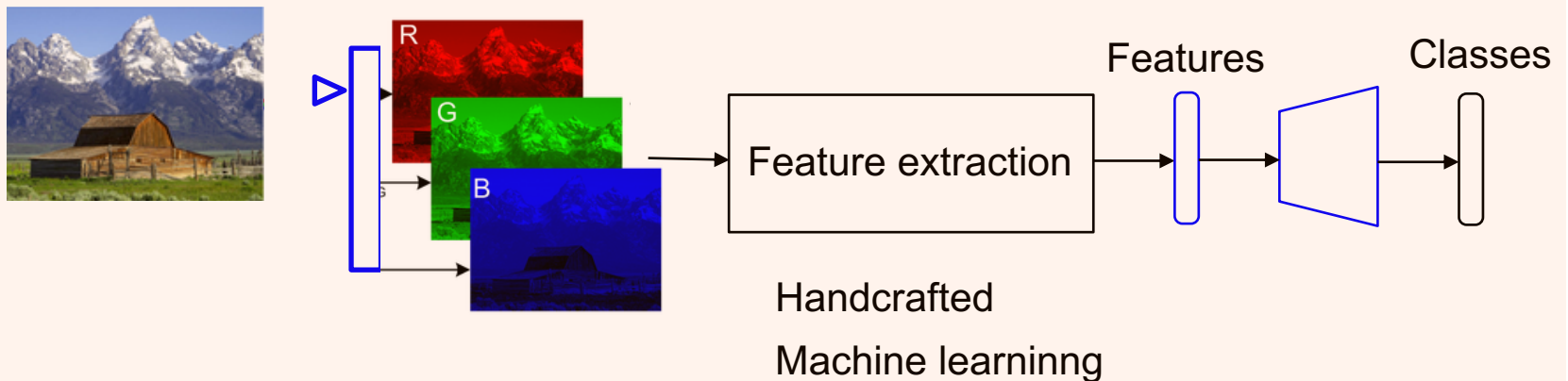
- “Hand-crafted”: designed under strict complexity/memory specs
- “Learned”: learned directly from data

Key techniques

- PCA/ICA
- Dimensionality reduction
- Decorrelation transforms
- Vector quantization (k-means)



Extraction



$$\text{ML: } \mathbb{Z} = f(x)$$

ML: $\mathbb{Z} = f_{\phi}(x)$ - learned function from data $\{x_i, c_i\}_{i=1}^N$

Example: deep learning

$$\mathbb{Z} = \sigma_L(W_L(\dots \sigma_1(W_1 x + b_1)))$$

$$\Phi = \{W_1, W_2, \dots, W_L; b_1, b_2, \dots, b_L\}$$

Mining

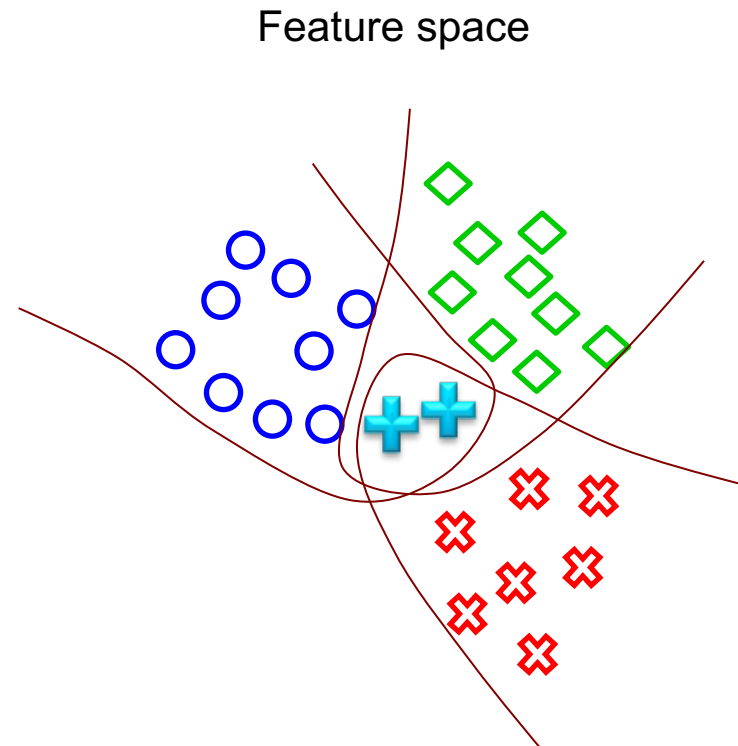
Common element(s) (or feature(s)) is unknown and to be discovered in the collection or database

Applications

- Images/Multimedia
- Biomarkers
- Market analysis

Basic techniques

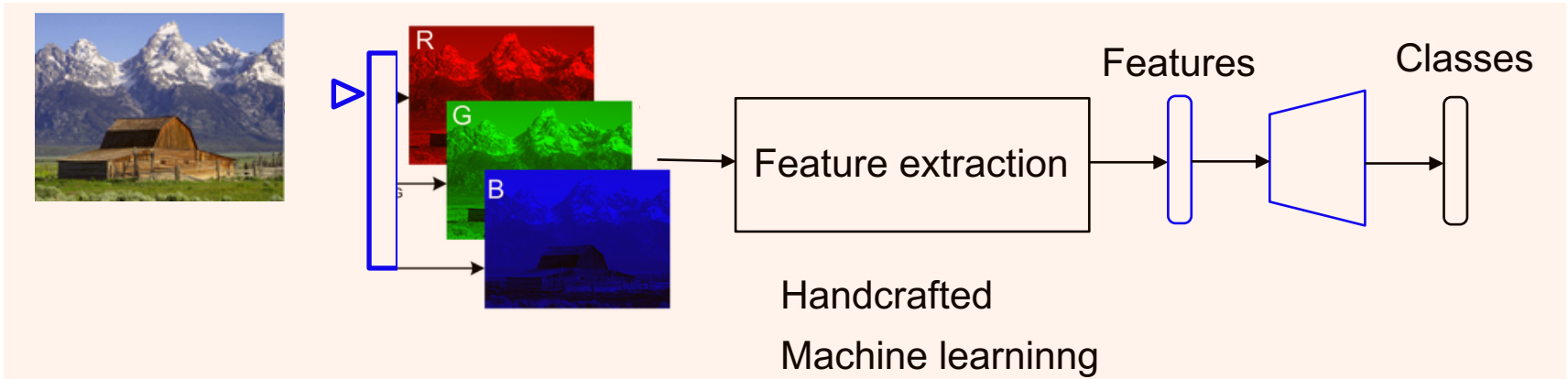
- Classification
- Clustering
- Prediction



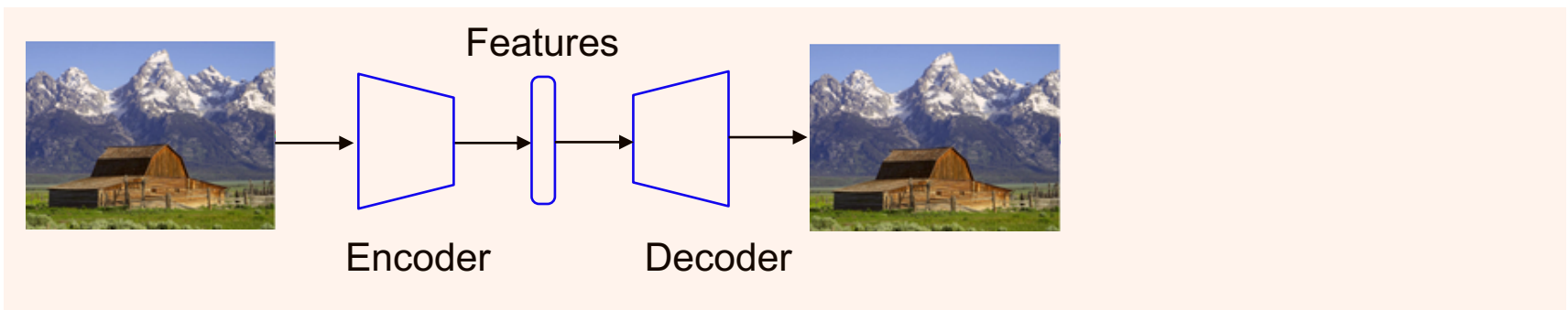
Mining

Typical architectures

Features taken from classification



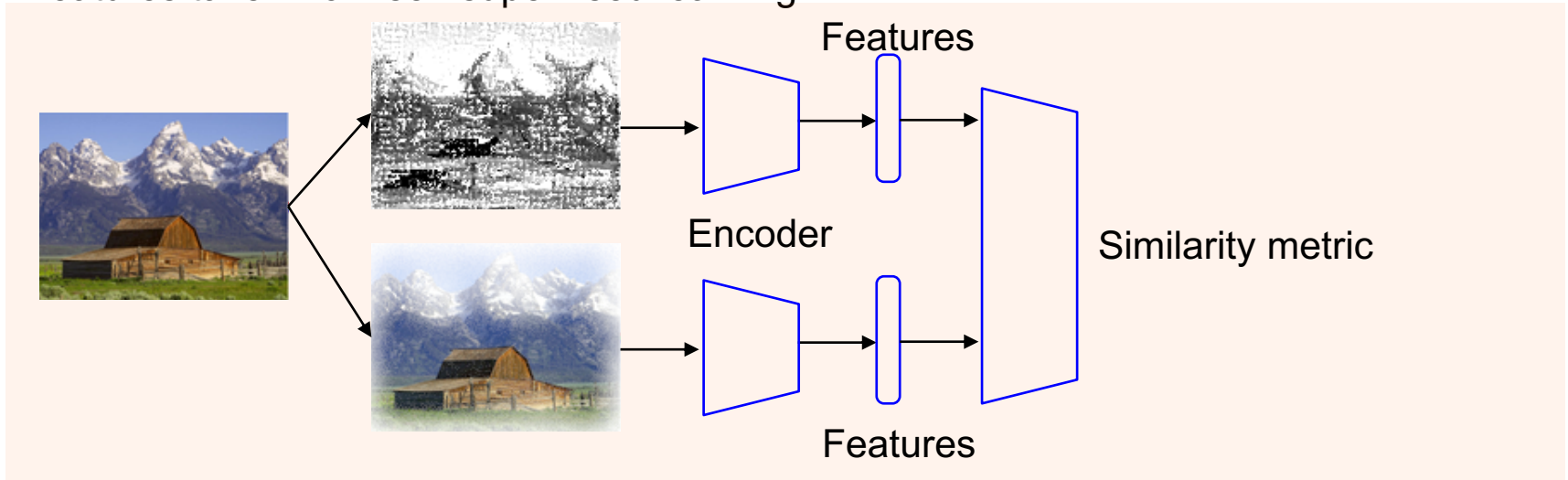
Features taken from auto-encoding



Mining

Typical architectures

Features taken from self-supervised learning



Recognition & Classification

Digital content/ Biometrics/ Physical objects

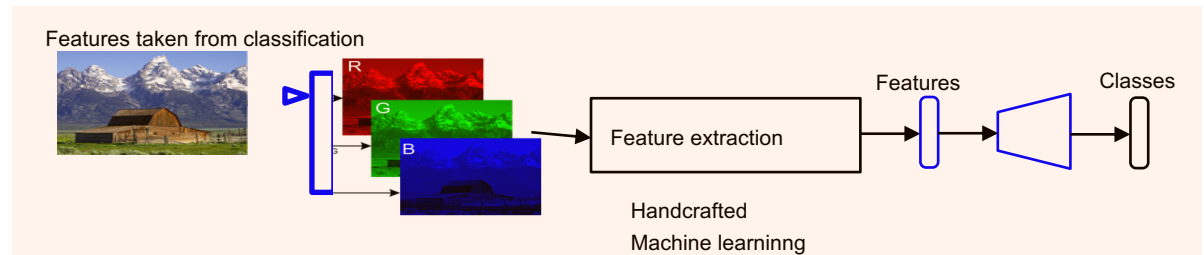
- Pattern recognition
- Information retrieval
- Classification
- Machine learning
- Identification/authentication

Common basis

Hypothesis testing

Two basic problems:

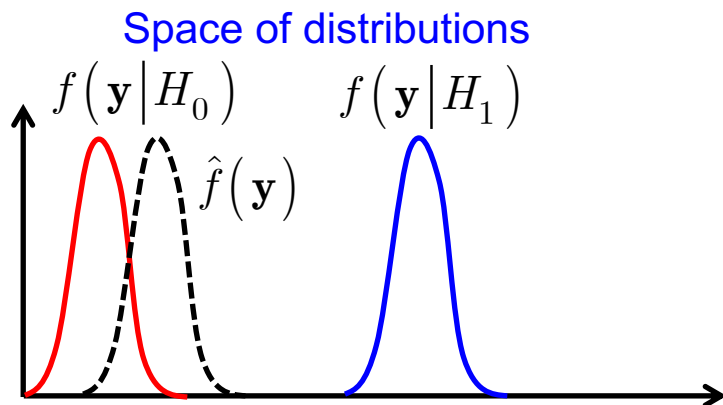
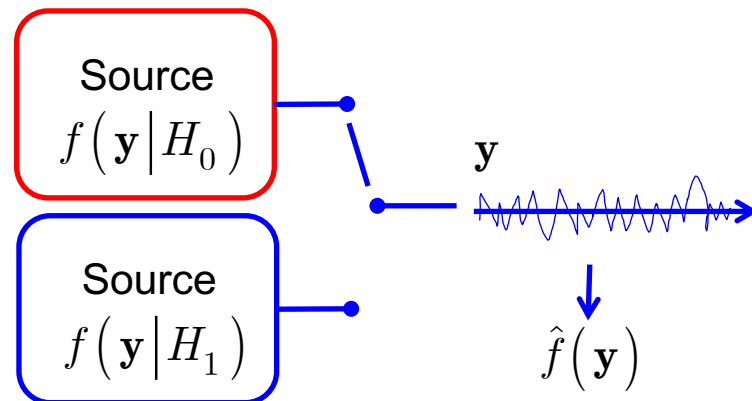
- 2-ary classification
- M-ary classification



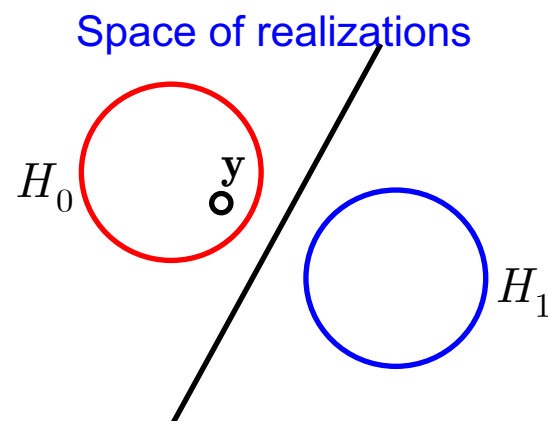
Applications: hypothesis testing

2-ary classification

Given: pdfs of classes $\begin{cases} H_0 : \mathbf{Y} \sim f(\mathbf{y}|H_0) \\ H_1 : \mathbf{Y} \sim f(\mathbf{y}|H_1) \end{cases}$
Decide: which class is observed



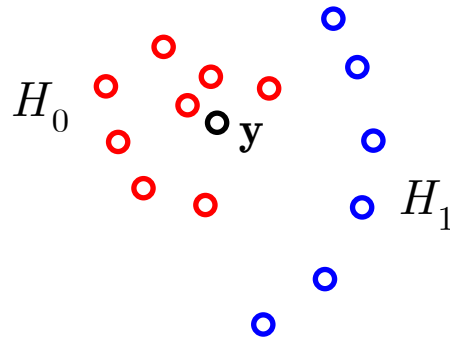
Main problem: measure the closeness of pdfs (under specific assumptions)



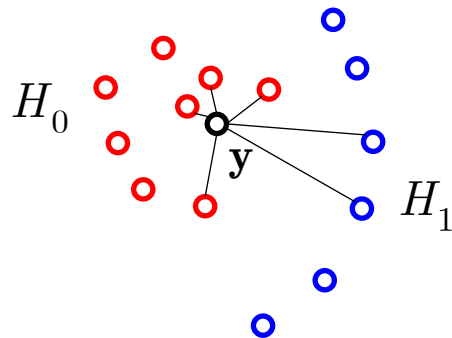
Main problem: measure the closeness of vectors

Applications: hypothesis testing

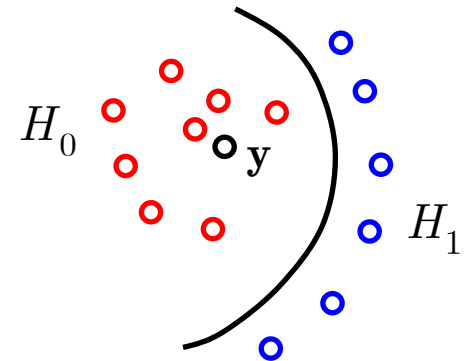
Space of realizations



k-NN classifier



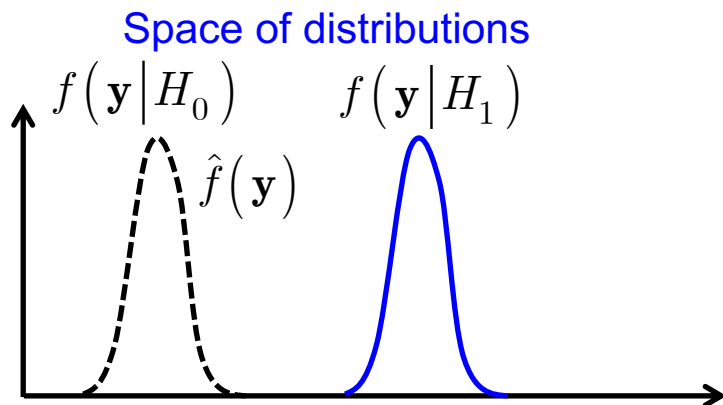
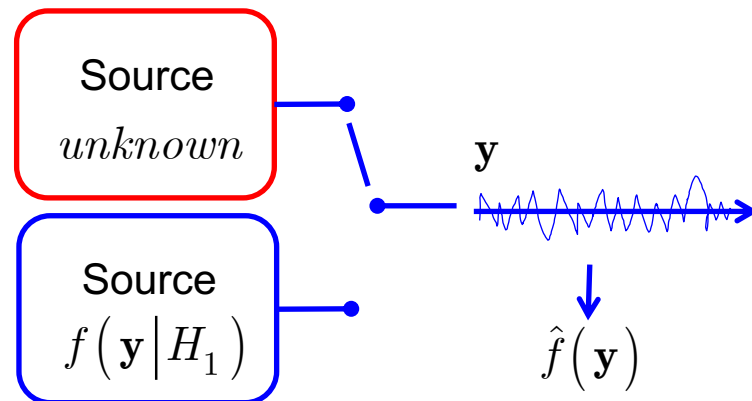
Deep classifier



Applications: hypothesis testing

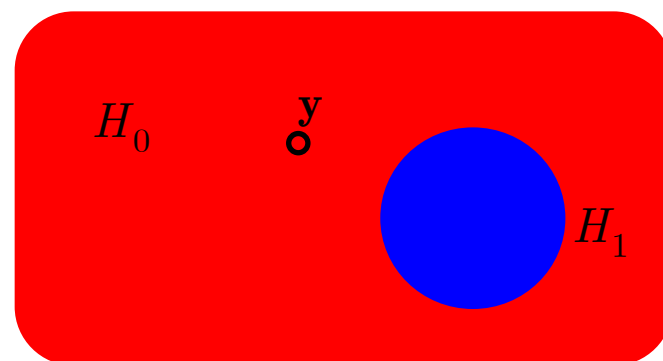
2-ary classification

Given: pdfs of classes $\begin{cases} H_0 : \mathbf{Y} \sim \text{any pdf} \\ H_1 : \mathbf{Y} \sim f(\mathbf{y}|H_1) \end{cases}$
Decide: which class is observed



Main problem: measure the closeness of pdfs

Space of realizations



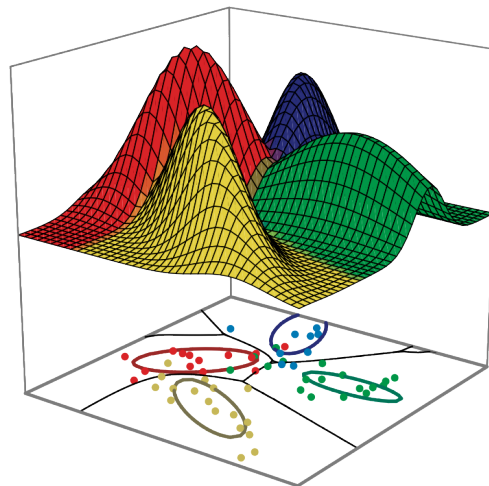
Main problem: measure in which region

Applications: hypothesis testing

M-ary classification

Given: pdfs of classes

Decide: which class is observed

$$\left\{ \begin{array}{l} H_0 : \mathbf{Y} \sim f(\mathbf{y} | H_0) \\ H_1 : \mathbf{Y} \sim f(\mathbf{y} | H_1) \\ \vdots \\ H_{M-1} : \mathbf{Y} \sim f(\mathbf{y} | H_{M-1}) \end{array} \right.$$


Main problem: efficient class separation

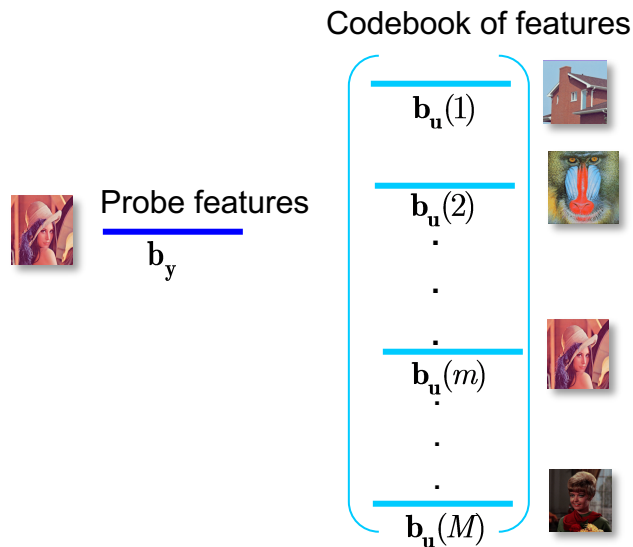
Applications: hypothesis testing

Hypothesis testing: link to machine learning

- theoretic setup: all pdfs of classes are known (IT)
 - practical setup: pdfs are not known but can be estimated
- Machine learning {
- **supervised learning:** labeled training data are provided
 - goal: to make predictions (estimate pdfs) based on examples (classify them, predict prices, etc)
 - **unsupervised learning:** unlabeled training data
 - goal: to reveal structure in the observed data (**data mining**)
 - **semi-supervised learning:** limited number of labeled data
 - goal: to improve predictions based on examples by making use of the additional “unlabeled” examples
 - **semi-supervised learning:** unlabeled training data

Search & Identification

Hypothesis testing: link to identification and retrieval



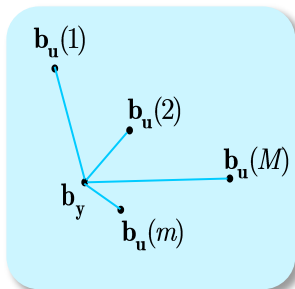
Question:

How many images M can be unequally ($P_e \rightarrow 0$) identified?

Answer:

M should be such that distortions do not cause any ambiguity

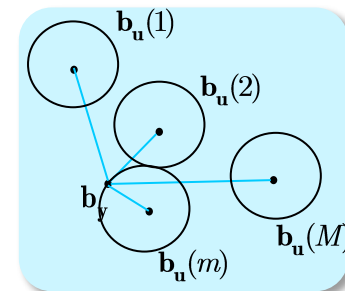
Consequence:



$$P_e = \frac{1}{2}P_f + \frac{1}{2}P_{ic}$$

P_f - probability of false acceptance

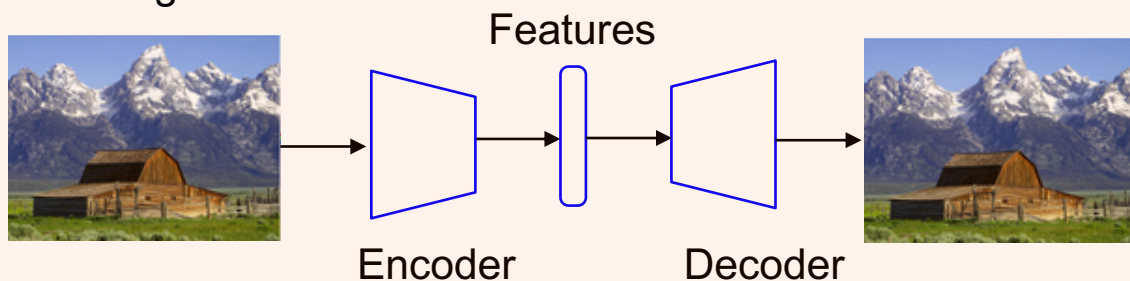
P_{ic} - probability of incorrect decoding



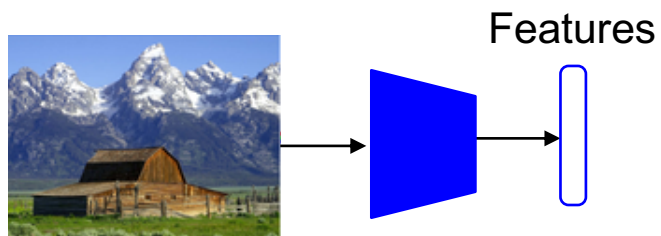
Search & Identification

Features taken from auto-encoding

Training

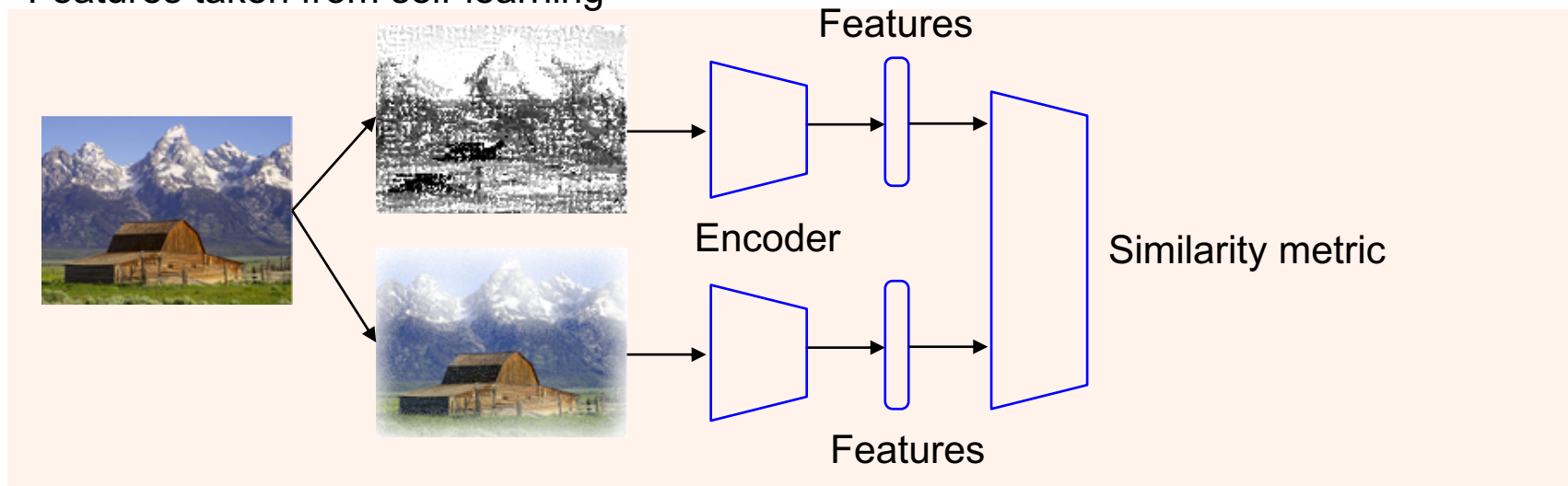


Embedding or feature extraction

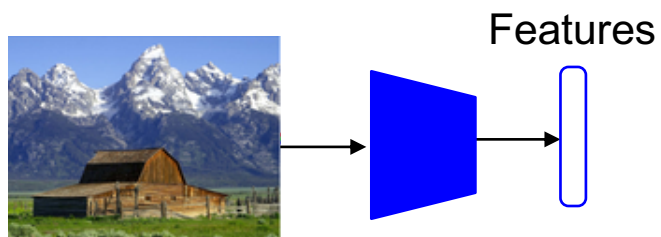


Search & Identification

Features taken from self-learning

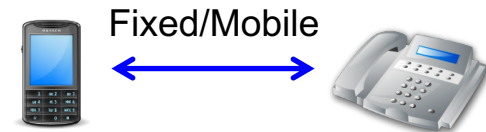
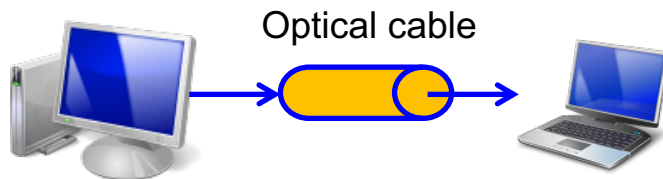
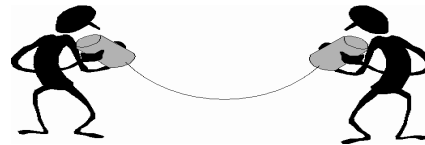


Embedding or feature extraction

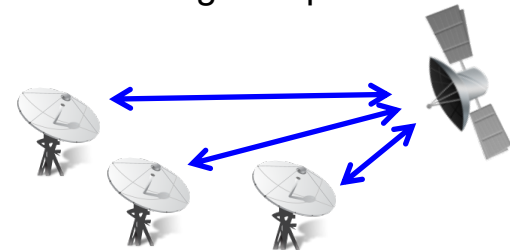


Transmission & Compression

Data transmission = digital communication



Broadcasting/multiple access systems



Transmission & Compression

